

COURSE NAME: COMPUTER NETWORKS  
COURSE CODE: CSA07

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#### COURSE OUTCOME COVERED

COS: Measure network performance using key metrics with simulation tools (M4.2B.2)

Assessment Tool Weightage: 30%

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Annexure A

### Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based

on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization on every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

#### RUBRICS FOR CALCULATION

| Criteria                  | Weightage | Level 4 – Exemplary                                                     | Level 3 – Proficient                        | Level 2 – Developing           | Level 1 – Beginning               |
|---------------------------|-----------|-------------------------------------------------------------------------|---------------------------------------------|--------------------------------|-----------------------------------|
| Problem Understanding     | 20%       | Identifies all inputs, outputs, constraints accurately 1                | Minor missing elements 2                    | Partial understanding 2        | Incorrect interpretation 2        |
| Algorithm Design          | 30%       | Complete, correct, and logically sound solution 2                       | Minor logical gaps 2                        | Partially correct logic 2      | Incorrect approach 2              |
| Use of Control Structures | 20%       | Efficient and appropriate use of loops, conditions, functions/modules 2 | Minor inefficiencies 2                      | Limited or basic usage 2       | Incorrect or missing constructs 2 |
| Pseudocode Structure      | 10%       | Well-structured, clear, consistent format, easy to follow 1             | Mostly clear with minor issues 2            | Difficult to follow in parts 1 | Poorly structured and unclear 2   |
| Completeness              | 20%       | Handles all cases, optimized approach 2                                 | Handles most cases, reasonable efficiency 2 | Handles basic cases only 1     | Incomplete or inefficient 1       |

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**COMPUTER NETWORKS  
PSEUDOCODE WRITING TASK  
REPORT**

Course Code: CSA07

Course Outcome Covered: CO5

Assessment Tool 2

Simulation Tools & Network Performance

**Complete Question & Answer Document**

Q1: In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.

Answer:

Algorithm Network\_Performance\_Evaluator

Inputs:

total\_data\_bits = 8000  
total\_time\_sec = 4  
packets\_sent = 200  
packets\_received = 180

BEGIN

// Step 1: Input Validation

IF total\_time\_sec == 0 OR packets\_sent == 0 THEN  
    PRINT "Error: Time or packets sent cannot be zero."  
    EXIT ALGORITHM

END IF

// Step 2: Calculate Network Metrics

throughput\_bps = total\_data\_bits / total\_time\_sec

packets\_lost = packets\_sent - packets\_received

packet\_loss\_percentage = (packets\_lost / packets\_sent) \* 100

// Step 3: Display Initial Results

PRINT "Calculated Network Throughput: " + throughput\_bps + " bps"

PRINT "Calculated Packet Loss: " + packet\_loss\_percentage + "%"

Step 4: Network Classification Logic

```
IF throughput_bps > 1500 AND packet_loss_percentage < 10 THEN  
    network_status = "Efficient"  
ELSE  
    network_status = "Inefficient"  
END IF
```

Step 5: Output Complete Summary

```
PRINT "-----"  
PRINT " Complete Network Performance Summary "  
PRINT "-----"  
PRINT "Total Data Received: " + total_data_bits + " bits"  
PRINT "Total Time: " + total_time_sec + " seconds"  
PRINT "Throughput: " + throughput_bps + " bps"  
PRINT "Packet Loss Percentage: " + packet_loss_percentage + "%"  
PRINT "Network Classification: " + network_status  
END
```

```
    PRINT "Status Link " & link_id & " operating normally."
  END IF
END FOR
```

```
  // Wait before initiating the next collection cycle
  WAIT(monitering_interval)
END WHILE
END
```

#### **Explanation:**

This algorithm significantly supports efficient bandwidth management by automating the constant real-time surveillance of all available network links. By continuously looping through the links and storing utilization data, administrators can track historical trends and traffic patterns. The condition that detects utilization exceeding 80% proactively identifies potential congestion before it leads to severe packet delays or drops. By instantly recommending and initiating a traffic switch to an alternate backup link, the algorithm guarantees seamless load balancing. This preventative strategy ensures that no single link becomes a bottleneck, maintaining smooth and optimized network communication.

```

IF path_status(path) == "FAILED" OR utilization > 100 THEN
    PRINT "Critical Alert Path: " + path.id + " - bandwidth/overload!"
    generate_administrative_alert(path)
    CONTINUE // Skip to evaluate next path
END IF

// Calculate comprehensive Link Quality Score
link_quality_score = (bandwidth * 0.5) + (delay * 0.3) + (utilization * 0.2)

// Identify the optimal path with highest score
IF link_quality_score > highest_quality_score THEN
    highest_quality_score = link_quality_score
    best_communication_path = path
END IF
END FOR

// Automatically re-route if optimal path changes
IF best_communication_path != current_active_path THEN
    PRINT "Network changed. Selecting new optimal path..."
    current_active_path = best_communication_path
    route_traffic(current_active_path)
    PRINT "Active Path: " + current_active_path.id
END IF

WAIT(10 seconds) // Interval before next collection
END WHILE
END

```

#### Explanation:

This comprehensive algorithm dramatically improves routing efficiency by calculating a dynamic Link Quality Score instead of relying on rigid shortest-path protocols. By integrating bandwidth, delay, and real-time utilization, it ensures traffic flows through the most capable routes. Network reliability and fault tolerance are strictly maintained by detecting failures instantly, issuing alerts, and automatically failing over to the next optimal path without human intervention. This proactive and continuous adaptation guarantees that communication remains stable and robust, successfully responding to fluctuating network conditions throughout the day.

Q3: A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Answer:

To meet the stringent security and compliance requirements of a financial institution without compromising network performance or vastly increasing the operational budget, a Zero-Trust Network Architecture based on logical network segmentation is absolutely critical. The core of this design relies on implementing Virtual Local Area Networks (VLANs) to segment the network logically rather than purchasing multiple expensive physical switches for each department. The network must be divided into strictly isolated VLANs: one for secure customer databases, one for general employee workstations, one for administrative management, and a Demilitarized Zone (DMZ) for external-facing online banking web servers.

To enforce strict access control boundaries between these segments, an internal Next-Generation Firewall (NGFW) should be deployed at the core of the network. The NGFW acts as a rigorous gateway, ensuring that a compromised employee workstation cannot directly communicate with the highly sensitive database VLAN. The NGFW will handle deep packet inspection (DPI) at wire speed, ensuring high throughput and negligible latency to preserve system performance. For continuous threat monitoring, an open-source or cost-effective Network